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Groove processing device and groove processing method

Abstract

A groove processing device that forms a groove in an object's surface using laser beams includes: a light source device that outputs the laser beams; a polygon mirror that reflects the laser beams that are output; a condensing optical system provided on an optical path of the laser beams that are reflected and focuses the laser beams; and a shielding plate provided between the condensing optical system and the object at a position which blocks some of the laser beams and blocks some of the laser beams. Among the laser beams that are focused, some that are not blocked form the groove in the surface of the object at a focus. The shielding plate is provided closer to the condensing optical system than the focus and is rotated with respect to the object's surface to block the laser beams that do not form the groove.

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Background/Summary

TECHNICAL FIELD OF THE INVENTION

(1) The present invention relates to a groove processing device and a groove processing method that form a groove in an object using a laser. The present application claims priority based on Japanese Patent Application No. 2019-091043 filed on May 14, 2019, the contents of which are incorporated herein by reference.

RELATED ART

(2) In the related art, a groove processing device is known which irradiate a surface of a steel sheet with a laser beam in a direction (scanning direction) intersecting a sheet travelling direction of the steel sheet, using a polygon mirror, to periodically form a groove in the surface of the steel sheet, thereby improving iron loss characteristics (see, for example, Patent Document 1).

PRIOR ART DOCUMENT

(3) [Patent Document]

(4) [Patent Document 1] Japanese Unexamined Patent Application, First Publication No. 2002-292484

DISCLOSURE OF THE INVENTION

Problems to be Solved by the Invention

(5) As shown in FIGS. 1A and 1B, a laser beam LB incident on a polygon mirror **10** of the groove processing device is not a point light source and has a predetermined radius ϕ .

(6) As shown in FIG. 1A, when the laser beam LB is incident so as to fall within one surface of the

polygon mirror **10**, the laser beam LB reflected by the polygon mirror **10** is focused on one spot on the surface of the steel sheet **20** through a condensing lens (hereinafter, simply referred to as a lens) **12**, and a groove is formed at the spot on the surface of the steel sheet **20**.

(7) On the other hand, as shown in FIG. **1B**, when the laser beam LB is incident on a corner portion in which two adjacent surfaces of the polygon mirror **10** meet, the laser beam LB is reflected from each of the two adjacent surfaces and is divided into two laser beams LB1 and LB2. The divided laser beams LB1 and LB2 are focused on the surface of the steel sheet **20** through the lens **12**. As a result, an end portion of the groove in the scanning direction is processed by the laser beams LB1 and LB2 with insufficient energy densities. Therefore, the end portion of the groove is shallow, and it is difficult to form a uniform groove. In addition, the divided laser beams LB1 and LB2 are irradiated in a direction different from that of the laser beam LB. Therefore, there is a concern that a position different from the position where a groove is to be formed in the surface of the steel sheet **20** or devices and the like other than the surface of the steel sheet **20** will be erroneously processed.

(8) In order to avoid this situation, a configuration is considered in which a shielding plate, such as a mask, is provided such that a portion corresponding to the end portion of the groove in the surface of the steel sheet **20** is not irradiated with the laser beams LB1 and LB2. However, this configuration has a problem that the shielding plate is processed and optical components are contaminated by minute pieces of the shielding plate generated by the processing.

(9) The invention has been made in view of the above-mentioned problems, and an object of the invention is to provide a groove processing device and a groove processing method that suppress the contamination of optical components and achieve uniform groove processing and groove depth.

Means for Solving the Problem

(10) Means for solving the problems include the following aspects. (1) According to an embodiment of the invention, there is provided a groove processing device that forms a groove in a surface of an object using laser beams. The groove processing device includes: a light source device that outputs the laser beams; a polygon mirror that reflects the laser beams output from the light source device; a condensing optical system that is provided on an optical path of the laser beams reflected by the polygon mirror and focuses the laser beams; and a shielding plate that is provided between the condensing optical system and the object at a position which blocks some of the laser beams focused through the condensing optical system and blocks some of the laser beams. Among the laser beams focused through the condensing optical system, some of the laser beams that are not blocked by the shielding plate form the groove in the surface of the object at a focus of the laser beams. The shielding plate is provided closer to the condensing optical system than the focus and is rotated with respect to the surface of the object so as to block the laser beams that do not form the groove. (2) In the groove processing device according to (1), when an angle of the shielding plate with respect to the surface of the object is ψ and a critical angle which is a maximum angle at which the laser beam falls within one plane mirror of the polygon mirror is $\theta_c(^{\circ})$, the angle ψ of the shielding plate may be inclined in a range of $2\theta_c < \psi \leq 90(^{\circ})$. (3) In the groove processing device according to (2), assuming that a position where a perpendicular line is drawn from a rotation axis of the polygon mirror to the plane mirror of the polygon mirror is a reference position, an angle formed between a boundary between two adjacent plane mirrors of the polygon mirror and the reference position is $\theta_0(^{\circ})$, a position where the shielding plate, which is inclined at the angle ψ , is irradiated with the laser beam reflected by the polygon mirror at an angle of $2\theta_0(^{\circ})$ when a rotation angle of the polygon mirror is $\theta_0(^{\circ})$ is a point P0, a position where the shielding plate, which is inclined at the angle ψ , is irradiated with the laser beam reflected by the polygon mirror at an angle of $2\theta_c(^{\circ})$ when the rotation angle of the polygon mirror is $\theta_c(^{\circ})$ is a point P, a height difference between the point P and the point P0 is L_{p0} , and a distance from the condensing optical system to a height of the point P is L_2 , $L_{p0} < L_2$ may be satisfied. (4) The groove processing device according to any one of (1) to (3) may further include: a position

adjustment portion that adjusts a position of the shielding plate in a scanning direction in which scanning is performed with the laser beams by the polygon mirror. (5) The groove processing device according to any one of (1) to (4) may further include: a housing that has the shielding plate disposed in a lower portion. The housing may have an upper opening portion that is located on the optical path of the laser beams focused by the condensing optical system, and a colorless and transparent window plate that transmits the laser beams without absorbing or reflecting the laser beams may be attached to the upper opening portion. (6) According to an embodiment of the invention, there is provided a groove processing method that forms a groove in a surface of an object using laser beams. The groove processing method includes: an output step of outputting the laser beams from a light source device; a reflection step of reflecting the laser beams output from the light source device by a polygon mirror; a condensing step of focusing the laser beams on the surface of the object using a condensing optical system that is provided on an optical path of the laser beams reflected by the polygon mirror; and a shielding step of blocking some of the laser beams using a shielding plate that is provided between the condensing optical system and the object at a position which blocks some of the laser beams focused through the condensing optical system. Among the laser beams focused through the condensing optical system, some of the laser beams that are not blocked by the shielding plate form the groove in the surface of the object at a focus of the laser beams. In the shielding step, the shielding plate is provided closer to the condensing optical system than the focus and is rotated with respect to the surface of the object so as to block the laser beams that do not form the groove. (7) In the groove processing method according to (6), in the shielding step, when an angle of the shielding plate with respect to the surface of the object is ψ and a critical angle which is a maximum angle at which the laser beam falls within one plane mirror of the polygon mirror is $\theta_c(^{\circ})$, the angle ψ of the shielding plate may be inclined in a range of $2\theta_c < \psi \leq 90(^{\circ})$. (8) In the groove processing method according to (7), in the shielding step, assuming that a position where a perpendicular line is drawn from a rotation axis of the polygon mirror to the plane mirror of the polygon mirror is a reference position, an angle formed between a boundary between two adjacent plane mirrors of the polygon mirror and the reference position is $\theta_0(^{\circ})$, a position where the shielding plate, which is inclined at the angle ψ , is irradiated with the laser beam reflected by the polygon mirror at an angle of $2\theta_0(^{\circ})$ when a rotation angle of the polygon mirror is $\theta_0(^{\circ})$ is a point P0, a position where the shielding plate, which is inclined at the angle ψ , is irradiated with the laser beam reflected by the polygon mirror at an angle of $2\theta_c(^{\circ})$ when the rotation angle of the polygon mirror is $\theta_c(^{\circ})$ is a point P, a height difference between the point P and the point P0 is L_{p0} , and a distance from the condensing optical system to a height of the point P is L_2 , $L_{p0} < L_2$ may be satisfied. (9) The groove processing method according to any one of (6) to (8) may further include: a shielding plate position adjustment step of adjusting a position of the shielding plate in a scanning direction in which scanning is performed with the laser beams by the polygon mirror. (10) The groove processing method according to any one of (6) to (9) may further include: a housing attachment step of attaching a colorless and transparent window plate that transmits the laser beams without absorbing or reflecting the laser beams to an upper opening portion of a housing that has the shielding plate disposed in a lower portion and has the upper opening portion which is located on the optical path of the laser beams focused by the condensing optical system.

Effects of the Invention

(11) According to the invention, the shielding plate is inclined to reduce the damage of the shielding plate which occurs when the shielding plate blocks the laser beam. Therefore, it is possible to provide a groove processing device and a groove processing method that suppress the contamination of optical components and achieve uniform groove processing and groove depth.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a schematic diagram showing a state in which a laser beam reflected from a polygon mirror is focused on a surface of a steel sheet when the laser beam is incident so as to fall within one surface of the polygon mirror.
- (2) FIG. 1B is a schematic diagram showing a state in which the laser beam reflected from each of two adjacent surfaces is focused on the surface of the steel sheet when the laser beam is incident across the two adjacent surfaces of the polygon mirror.
- (3) FIG. 2 is a schematic diagram showing a configuration of a groove processing device according to an embodiment of the invention as viewed from a rolling direction of the steel sheet.
- (4) FIG. 3 is a schematic diagram showing a rotation angle of the polygon mirror.
- (5) FIG. 4 is a schematic diagram showing a configuration of a movable shielding plate device.
- (6) FIG. 5 is a schematic diagram showing the optimum position and angle of a shielding plate.
- (7) FIG. 6 is a graph showing a relationship between an angle ψ of the shielding plate and a height difference $Lp0$.

EMBODIMENTS OF THE INVENTION

- (8) Hereinafter, an embodiment of the invention will be described with reference to the drawings. In the specification and the drawings, the same components are designated by the same reference numerals.
- (9) FIG. 2 schematically shows a configuration of a groove processing device **100** according to the embodiment of the invention as viewed from a rolling direction of a steel sheet **20**. The groove processing device **100** is a device that periodically forms a groove in a surface of the steel sheet **20**, which is an object to be processed, using a laser. The steel sheet **20** is made of, for example, a well-known grain-oriented electrical steel sheet material. In the groove processing device **100**, the position of the steel sheet **20** in a width direction is set on the basis of the length and position of the groove formed in the surface of the steel sheet **20**, and the position of the steel sheet **20** in a longitudinal direction is set on the basis of the dimensions of the groove processing device **100**. The width direction of the steel sheet **20** is a scanning direction of the laser and is a left-right direction of the plane of paper in FIG. 2. The longitudinal direction of the steel sheet **20** is the rolling direction of the steel sheet **20** and is a depth direction of the plane of paper in FIG. 2.
- (10) As shown in FIG. 2, the groove processing device **100** includes a polygon mirror **10**, a light source device **11**, a collimator **11A**, a lens **12**, and a movable shielding plate device **30**.
- (11) The polygon mirror **10** has, for example, a regular polygonal prism shape, and a plurality of (N) plane mirrors are provided on each of a plurality of side surfaces constituting a regular polygonal prism. A laser beam LB is incident on the polygon mirror **10** from the light source device **11** through the collimator **11A** in one direction (horizontal direction) and is then reflected by the plane mirror (reflection step).
- (12) The polygon mirror **10** has a configuration in which it can be rotated on a rotation axis O1 by the driving of a motor (not shown), and the incident angle of the laser beam LB on the plane mirror changes sequentially depending on the rotation angle of the polygon mirror **10**. Therefore, the polygon mirror **10** sequentially changes the reflection direction of the laser beam LB such that the steel sheet **20** is scanned with the laser beam LB in the width direction.
- (13) In addition, FIGS. 1A, 1B, 2, and 3 show an example in which the polygon mirror **10** has eight plane mirrors. However, the number of plane mirrors constituting the polygon mirror **10** is not particularly limited.
- (14) The light source device **11** outputs a laser beam using a predetermined irradiation method (for example, a continuous irradiation method or a pulse irradiation method) under the control of a control unit (not shown) (output step).
- (15) The collimator **11A** is connected to the light source device **11** through an optical fiber cable **15**. The collimator **11A** adjusts the radius of the laser beam output from the light source device **11** and

outputs the adjusted laser beam LB to the polygon mirror **10**. The laser beam LB output to the polygon mirror **10** has a laser diameter having a predetermined radius ϕ , and the laser diameter is that of a circle. However, the laser diameter may be that of an ellipse. In this case, an elliptical condensing shape can be formed by inserting a cylindrical lens or a cylindrical mirror between the collimator **11A** and the polygon mirror **10** to change the radius of the beam along one axis (for example, a scanning direction).

(16) The lens **12** is a condensing optical system that is provided on the optical path of the laser beam reflected by the polygon mirror **10** and is manufactured by performing processing, such as grinding and polishing, on a piece of glass. In addition, a mirror may be adopted as the condensing optical system constituting the groove processing device **100** instead of the condensing lens **12**.

(17) The lens **12** may have a non-condensing portion (not shown) that is integrally provided outside (in the outer circumference of) the lens **12**. The non-condensing portion is located on the optical paths of laser beams LB1 and LB2 that have been divided and reflected from a corner portion in which two adjacent plane mirrors of the polygon mirror **10** meet and transmits the divided laser beams LB1 and LB2. The non-condensing portion is a planar optical system of a donut-shaped plate. The non-condensing portion does not have a focus because the focal length thereof is infinite. Since the laser beams LB1 and LB2 that have passed through the non-condensing portion are not focused, they do not have a high energy density. Therefore, even when a shielding plate **35** is irradiated with the laser beams LB1 and LB2 that have passed through the non-condensing portion, the damage of the shielding plate **35** is small. In addition, the non-condensing portion may not be the planar optical system and may be, for example, an optical system that diverges the divided laser beams LB1 and LB2.

(18) The movable shielding plate device **30** which will be described below is provided between the lens **12** and the steel sheet **20**. The movable shielding plate device **30** is disposed on the optical path of the laser beam LB that is reflected by the polygon mirror **10** and passes through the lens **12**. The laser beam LB reflected by the polygon mirror **10** passes through the lens **12** and the movable shielding plate device **30** and is focused on the surface of the steel sheet **20** (condensing step). Therefore, a groove is formed in the surface of the steel sheet **20**.

(19) Further, in a groove processing method which irradiates the surface of the steel sheet **20** with the laser beam LB to form a groove, base steel sheet is melted and removed to form a groove. Therefore, as the groove becomes deeper, the probability that a molten protrusion will occur on the surface becomes higher. Therefore, the groove processing device **100** may be configured to include a supply nozzle (not shown) which injects an assist gas for blowing off a molten material and is provided at a predetermined position. Further, the collimator **11A**, the polygon mirror **10**, the lens **12**, and the movable shielding plate device **30** of the groove processing device **100** may be covered with a housing (not shown), and the inside of the housing may be filled with a clean gas such that the internal pressure of the housing is positive. In this case, it is possible to prevent a molten material and the like from entering the housing and to prevent the optical system of the groove processing device **100** from being contaminated by the molten material and the like.

(20) Next, the rotation angle of the polygon mirror **10** will be described with reference to FIG. 3. In this embodiment, it is assumed that the rotation angle $\theta(^{\circ})$ of the polygon mirror **10** is defined by a central angle with respect to a reference position for each of the plane mirrors constituting the polygon mirror **10**. As shown in FIG. 3, it is assumed that a position where a perpendicular line PL is drawn from the rotation axis O1 of the polygon mirror **10** to a plane mirror **101** is the reference position ($\theta=0(^{\circ})$). The rotation angle θ of the polygon mirror **10** is an angle (central angle) formed between the position of a center LBC of the laser beam LB incident on each plane mirror and the reference position ($\theta=0(^{\circ})$). In FIG. 3, a counterclockwise angle from the reference position ($\theta=0(^{\circ})$); the perpendicular line PL) is defined as a positive angle, and a clockwise angle from the reference position is defined as a negative angle.

(21) An angle θ_0 formed between the reference position ($\theta=0(^{\circ})$) in each plane mirror and a

boundary with an adjacent plane mirror is $180(^{\circ})/N$. The rotation angle θ of one plane mirror is defined in the range of $-\theta_0 \leq \theta \leq +\theta_0$. Therefore, in FIG. 3, the rotation angle $\theta = +\theta_0$ of the plane mirror **101** and the rotation angle $\theta = -\theta_0$ of a plane mirror **102** adjacent to the plane mirror **101** in the counterclockwise direction indicate the same position on the polygon mirror **10**.

(22) In this embodiment, the maximum angle at which the incident laser beam LB falls within one surface (one plane mirror) of the polygon mirror **10** is defined as a critical angle θ_c . That is, when the laser beam LB is totally reflected by one plane mirror without being divided by a corner portion in which two adjacent plane mirrors of the polygon mirror **10** meet, the critical angle θ_c is the maximum angle at which the center LBC of the laser beam LB is located. Assuming that the radius (circumscribed radius) of a circumscribed circle Cl of the polygon mirror **10** is R and the radius of the laser beam LB incident on the polygon mirror **10** is ϕ , the critical angle θ_c is defined by the following Expression (1).

$$(23) \quad c = \sin^{-1} [(R \times \sin \theta_0 - \phi) / R] \quad (1)$$

(24) Next, a specific configuration of the movable shielding plate device **30** will be described with reference to FIG. 4. As shown in FIG. 4, the movable shielding plate device **30** has a configuration in which it has a box-shaped housing **31** formed of, for example, a metal material and a pair of shielding plates **35** disposed so as to face each other in a scanning direction x of the laser beam LB are disposed in a lower portion of the housing **31**. In addition, the shielding plate **35** is rotated on a rotating portion **37a** as a fulcrum, which will be described below. In FIG. 4, **35a** indicates the shielding plate when the shielding plate **35** is inclined by rotation.

(25) In the housing **31**, an upper opening portion **31a** is formed in an upper portion that is located on the optical path of the laser beam LB focused by the lens **12**, a lower opening portion **31b** is formed in a lower portion, and a colorless and transparent window plate **33** is attached to the upper opening portion **31a** (housing attachment step). The window plate **33** is, for example, a glass plate. The window plate **33** transmits the laser beam without absorbing or reflecting the laser beam. For example, the window plate **33** is obtained by coating both surfaces of a synthetic quartz glass plate with antireflection films. Therefore, the upper opening portion **31a** can be closed by the window plate **33**, and the laser beam LB that has been reflected by the polygon mirror **10** and passed through the lens **12** can pass through the upper portion of the housing **31**.

(26) Further, the laser beam LB which has passed through the window plate **33** from the lens **12** passes through the lower opening portion **31b** of the housing **31**, and the surface of the steel sheet **20** is irradiated with the laser beam LB. When the polygon mirror **10** is rotated, the inclination angle of the laser beam LB changes depending on the rotation angle of the polygon mirror **10**. The irradiation position of the laser beam LB passing through the movable shielding plate device **30** is moved on the surface of the steel sheet **20** in the width direction of the steel sheet **20**. That is, the laser beam LB passing through the movable shielding plate device **30** is moved on the surface of the steel sheet **20** in the width direction of the steel sheet **20** as the scanning direction x.

(27) In the lower opening portion **31b** of the housing **31**, the shielding plates **35** are provided in the vicinity of opening portion ends **31c** which are disposed so as to face each other in the scanning direction x of the laser beam LB. The shielding plate **35** is provided between the lens **12** and the steel sheet **20**. That is, the shielding plate **35** is provided closer to the lens **12** than the focus of the laser beam LB that has passed through the lens **12**. The shielding plate **35** blocks some of the laser beams focused through the lens **12** (shielding step). The pair of shielding plates **35** disposed so as to face each other in the scanning direction x of the laser beam LB have the same configuration and are formed of, for example, a steel material in a plate shape. Each of the shielding plates **35** is provided with a position adjustment portion **34** that adjusts the position of the shielding plate **35** in the scanning direction x of the laser beam LB and an angle adjustment portion **36** that adjusts the angle of a plate surface of the shielding plate **35** with respect to the surface of the steel sheet **20**.

(28) In this embodiment, the position adjustment portion **34** is, for example, a guide groove that is

formed in the lower opening portion **31b** of the housing **31**, and the guide groove is formed along the scanning direction x of the laser beam **LB**. The position adjustment portion **34** is provided such that the shielding plate **35** can be slid in the guide groove and moves the shielding plate **35** along the guide groove in the scanning direction x of the laser beam **LB** (shielding plate position adjustment step).

(29) As described above, the position adjustment portion **34** adjusts the position of the shielding plate **35** in the scanning direction x of the laser beam **LB** such that the shielding plate **35** is irradiated with the laser beam **LB** moved to the end portion of the steel sheet **20** in the width direction when the laser beam **LB** that has passed through the lens **12** is moved from the center to the end portion of the steel sheet **20** along the scanning direction x . This configuration makes it possible to adjust the range of the scanning direction x (the width direction of the steel sheet **20**) in which the shielding plate **35** is irradiated with the laser beam **LB**.

(30) The shielding plate **35** blocks some of the laser beams **LB** that are focused through the lens **12** and moved in the scanning direction x at the end in the scanning direction such that the surface of the steel sheet **20** is not irradiated with the laser beam **LB** at the end in the scanning direction and no grooves are formed in the surface of the steel sheet **20**. On the other hand, among the laser beams **LB** which are focused through the lens **12** and moved in the scanning direction x , the remaining laser beams **LB** which are not blocked by the shielding plate **35** converge into the focus of the laser beams **LB** on the surface of the steel sheet **20** to form a groove.

(31) Therefore, the shielding plate **35** blocks unnecessary beams which are irradiated with positions other than a groove processing position on the steel sheet **20** among the laser beams **LB** which have been reflected by the plane mirror of the rotating polygon mirror **10** and have a high energy density or the laser beams **LB1** and **LB2** which have been divided in the corner portion of the polygon mirror **10** and have a low energy density. Therefore, the end portion of the groove in the scanning direction x is not shallow, and positions other than the position of the groove on the steel sheet **20** are not processed. As a result, it is possible to achieve uniform groove processing and groove depth in the steel sheet **20**.

(32) In this embodiment, the angle adjustment portion **36** includes the rotating portion **37a** that enables the shielding plate **35** to rotate with respect to the lower portion of the housing **31**, a guide portion **37b** that defines a trajectory on which the shielding plate **35** is inclined, and a connection portion **37c** that rotatably connects the shielding plate **35** to the guide portion **37b**. The shielding plates **35** are rotated such that the surfaces of the shielding plates **35** face each other. Therefore, the rotating portion **37a** rotates the shielding plate **35** on a rotation axis to incline a flat sheet surface of the shielding plate **35** with respect to the surface of the steel sheet **20**.

(33) In the case of this embodiment, the rotating portion **37a** and the guide portion **37b** are provided so as to be slidable along the scanning direction of the laser beam **LB** by the position adjustment portion **34** in operative association with the shielding plate **35**. That is, the rotating portion **37a** has a configuration in which it is provided in, for example, the position adjustment portion **34** and is movable along the scanning direction of the laser beam **LB** by the sliding of the position adjustment portion **34**. In addition, the guide portion **37b** is, for example, a guide groove that is provided in a part of the position adjustment portion **34** along the inner wall of the housing **31** and has a configuration in which it is movable along the scanning direction by the sliding of the position adjustment portion **34** and guides the trajectory of the connection portion **37c** such that the shielding plate **35** having the connection portion **37c** can be moved along the scanning direction of the laser beam **LB**.

(34) Here, the guide portion **37b** is, for example, an annular member that is made of a metal material or the like and has a curved elongated hole, and the connection portion **37c** provided in the shielding plate **35** is moved along the curved elongated hole. In this case, the connection portion **37c** is, for example, a protrusion member that is disposed in the elongated hole of the guide portion **37b** and is moved along the elongated hole of the guide portion **37b**. The shielding plate **35** is

rotated on the rotating portion **37a** that is provided in the lower end portion of the shielding plate **35**, and the connection portion **37c** is moved along the curved elongated hole of the guide portion **37b** to change the inclination angle of the shielding plate **35** with respect to the surface of the steel sheet **20**.

(35) The shielding plate **35** is rotated with respect to the surface of the steel sheet **20** so as to block the laser beam LB that does not form a groove. Therefore, the angle adjustment portion **36** adjusts the angle of the shielding plate **35** with respect to the steel sheet **20** to prevent the shielding plate **35** from being irradiated with the laser beam LB having a high energy density when the shielding plate **35** is irradiated with the laser beams LB. Therefore, it is possible to reduce the damage of the shielding plate **35** caused by irradiation with the laser beam LB.

(36) That is, the laser beam LB has a laser diameter having a predetermined radius ϕ . However, the size of the laser diameter that appears on the shielding plate **35** changes due to a change in the angle of the shielding plate **35** with respect to the laser beam LB, which causes a change in the energy density of the laser beam LB irradiated with the shielding plate **35**. Therefore, it is possible to set the shielding plate **35** to an angle at which the damage of the shielding plate **35** is reduced.

(37) Further, the laser beam LB is focused by the lens **12** so as to have the highest energy density on the surface of the steel sheet **20** on which the laser beam LB is focused. Therefore, an object that is at the same distance (that is, a focal length) as the distance from the lens **12** to the steel sheet **20** is irradiated with the laser beam LB having a high energy density. As the distance from the lens increases, the energy density decreases. Therefore, the shielding plate **35** that is provided closer to the lens **12** than the focus is inclined to balance the distance from the lens **12** and the change in the laser diameter, which makes it possible to obtain a suitable energy density and to reduce the damage of the shielding plate **35**.

(38) Next, the position of the shielding plate **35** in the scanning direction of the laser beam LB will be described with reference to FIG. 5. In addition, in FIG. 5, **10a** indicates a portion of the plane mirror when the polygon mirror **10** is rotated. In this case, it is assumed that the focal length of the lens **12** which is the condensing optical system is f . When the polygon mirror **10** is rotated by $\theta(^{\circ})$, the laser beam LB reflected by the polygon mirror **10** is moved by $2\theta(^{\circ})$. Then, when the rotation angle $\theta(^{\circ})$ of the polygon mirror **10** is from $\theta c(^{\circ})$ to $\theta 0(^{\circ})$, the laser beam LB needs to be blocked by the shielding plate **35**.

(39) Here, it is assumed that a distance from the plane mirror of the polygon mirror **10** to the lens **12** which is the condensing optical system is $L1$. Further, it is assumed that a distance from the lens **12** which is the condensing optical system to the height of a point P at the position where the shielding plate **35** is irradiated with the laser beam LB reflected by the polygon mirror **10** at an angle of $2\theta c(^{\circ})$ when the rotation angle θ of the polygon mirror **10** is $\theta c(^{\circ})$ is $L2$. In addition, in this embodiment, as shown in FIG. 5, the lower end portion of the shielding plate **35** in which the rotating portion **37a** is provided is the point P.

(40) Furthermore, it is assumed that a perpendicular line which is drawn from the polygon mirror **10** to the steel sheet **20** and through which the laser beam LB passes when the rotation angle θ of the polygon mirror **10** is $0(^{\circ})$ is PL1. Moreover, it is assumed that a straight line which horizontally extends from the point P at the position where the shielding plate **35** is irradiated with the laser beam LB reflected by the polygon mirror **10** when the rotation angle θ of the polygon mirror **10** is $\theta c(^{\circ})$ to the perpendicular line PL1 is XL1. Then, assuming that the point where the perpendicular line PL1 through which the laser beam LB passes and the straight line XL1 from the point P intersect each other is P1, a distance d between the point P and the point P1 can be represented by the following Expression (2). In addition, as described above, ϕ indicates the radius of the laser beam LB incident on the polygon mirror **10** (FIG. 3).

$$(41) \quad d = (L1 + L2) \times \tan 2\theta c - \left\{ \phi / f \times (f - L2) / \cos 2\theta c \right\} \quad (2)$$

(42) In this embodiment, the position adjustment portion **34** moves the shielding plate **35** in the

scanning direction of the laser beam LB to adjust the distance d from the perpendicular line PL1 through which the laser beam LB passes to the position of the shielding plate 35 to the distance d calculated by the above-mentioned Expression (2).

(43) Next, an angle ψ of the shielding plate 35 with respect to the surface of the steel sheet 20 will be described with reference to FIG. 5. In this case, the angle ψ of the shielding plate 35 is an angle formed between the surface direction of the surface of the steel sheet 20 and the plate surface of the shielding plate 35. Here, the energy density of the laser beam LB is highest at the point P of the shielding plate 35 when the angle ψ of the shielding plate 35 with respect to the surface of the steel sheet 20 is $20c(^{\circ})$ at which the laser beam LB is vertically incident on the shielding plate 35.

(44) Therefore, it is desirable to incline the shielding plate 35 as much as possible while avoiding that the angle ψ of the shielding plate 35 is close to $20c(^{\circ})$ in order to avoid the damage of the surface of the shielding plate 35 by the laser beam LB as much as possible.

(45) Assuming that the energy density of the laser beam LB when the angle ψ of the shielding plate 35 is $\psi=20c$ is I_{pc} , the energy density I_{pc} of the laser beam LB can be represented by the following Expression (3). In addition, P indicates the laser power (W) of the laser beam LB.

$$(46) \quad I_{pc} = P / \{ \pi \times (f - L/2)^2 \} \quad (3)$$

(47) The energy density I_p of the laser beam LB when the shielding plate 35 is inclined at the angle ψ can be represented by the following Expression (4).

$$(48) \quad I_p = I_{pc} \times \cos(\psi - 20c) \quad (4)$$

(49) When the angle ψ of the shielding plate 35 is from $0(^{\circ})$ to $20c(^{\circ})$, the focusing diameter of the laser beam LB is small, and the energy density I_p of the laser beam LB is high in a portion other than the point P, which is not desirable. In addition, when the angle ψ of the shielding plate 35 is $90(^{\circ})$ or greater, the shielding plate 35 is inclined too much, and the influence of the laser beam LB irradiated with the side surface of an upper portion of the shielding plate 35 is large. Therefore, unexpected processing is performed, which is not desirable.

(50) Therefore, it is desirable that the angle ψ of the shielding plate 35 is $20c < \psi \leq 90(^{\circ})$. In this embodiment, the angle adjustment portion 36 changes the inclination of the shielding plate 35 using the rotating portion 37a as the rotation axis to adjust the angle ψ of the shielding plate 35 with respect to the surface of the steel sheet 20 in the range of $20c < \psi \leq 90(^{\circ})$.

(51) When the angle ψ of the shielding plate 35 is large, the lens 12 which is the condensing optical system is close to the tip of the shielding plate 35. For this reason, it is not desirable that the angle ψ of the shielding plate 35 is too large. Therefore, it is more desirable that the angle ψ of the shielding plate 35 is $80(^{\circ})$ or less.

(52) Next, constraint conditions will be described. Here, it is assumed that, when the shielding plate 35 is inclined at the angle ψ , the position where the shielding plate 35 is irradiated with the laser beam LB reflected by the polygon mirror 10 at an angle of $200(^{\circ})$ when the rotation angle θ of the polygon mirror 10 is $0(^{\circ})$ is a point P0.

(53) Then, similarly, assuming that, when the shielding plate 35 is inclined at the angle ψ , the height difference between the point P at the position where the shielding plate 35 is irradiated with the laser beam LB reflected by the polygon mirror 10 at an angle of $20c(^{\circ})$ when the rotation angle θ of the polygon mirror 10 is $0c(^{\circ})$ and the point P0 is L_{p0} , L_{p0} can be represented by the following Expression (5).

$$(54) \quad L_{p0} = (L_1 + L_2) \times \tan 200 \times \sin \psi / \{ \sin(90 + 200 - \psi) \times \cos 200 \} \quad (5)$$

(55) Here, the height difference L_{p0} needs to be smaller than the distance L2 from the lens 12 which is the condensing optical system to the height of the point P. Therefore, $L_{p0} < L_2$ needs to be satisfied.

(56) In the above-mentioned configuration, in the groove processing device 100, the shielding plate 35 provided between the lens 12 and the steel sheet 20 blocks the laser beam LB that has passed through the lens 12, and the angle adjustment portion 36 inclines the shielding plate 35 at the angle

ψ with respect to the surface of the steel sheet **20** when an end portion of a groove is formed in the surface of the steel sheet **20**. As described above, in the groove processing device **100**, the shielding plate **35** is inclined at the angle ψ to reduce the damage of the shielding plate **35** which occurs when the shielding plate **35** blocks the laser beam LB. Therefore, it is possible to achieve uniform groove processing and groove depth, without contaminating optical components, and to produce a product having excellent iron loss characteristics.

(57) Further, in the groove processing device **100**, the position adjustment portion **34** adjusts the position of the shielding plate **35** in the scanning direction x in which scanning is performed with the laser beam LB by the polygon mirror **10**. Therefore, in the groove processing device **100**, when the laser beam LB that has passed through the lens **12** is moved from the center to the end portion of the steel sheet **20** along the scanning direction x , the shielding plate **35** is irradiated with the laser beam LB reflected by the plane mirror of the polygon mirror **10**. As a result, a groove that has a uniform groove depth even in an end portion can be formed in the surface of the steel sheet **20**. In addition, it is possible to adjust the range of the scanning direction x (the width direction of the steel sheet **20**) in which the shielding plate **35** is irradiated with the laser beam LB.

(58) Further, in the groove processing device **100**, when the angle ψ of the shielding plate **35** is adjusted, it is desirable to set the angle ψ in the range of $2\theta_c < \psi \leq 90(^{\circ})$ while satisfying $L_{p0} < L_2$ as a constraint condition. Furthermore, it is desirable that the most desirable angle ψ of the shielding plate **35** is in the range of $2\theta_c < \psi \leq 90(^{\circ})$ and in the angle range of the central angle $\pm 5(^{\circ})$ satisfying the constraint condition of $L_{p0} < L_2$. As described above, the angle ψ of the shielding plate **35** is adjusted to the range of $2\theta_c < \psi \leq 90(^{\circ})$ and to the angle range of the central angle $\pm 5(^{\circ})$ satisfying the constraint condition of $L_{p0} < L_2$ to more reliably reduce the damage of the shielding plate **35** by the laser beam LB.

(59) In addition, the shielding plate according to the above-described embodiment may be formed of a material that absorbs the laser beam LB. For example, a black alumite treatment or absorption lacquer coating is performed on the surface of the shielding plate in order for the shielding plate to absorb the energy of the laser beam. Further, a water channel may be provided in the shielding plate to perform indirect water cooling in order to cool the shielding plate.

(60) Furthermore, in the above-described embodiment, the groove processing device **100** provided with both the position adjustment portion **34** and the angle adjustment portion **36** has been described. However, the invention is not limited thereto, and the groove processing device may be provided with only the angle adjustment portion **36**.

(61) Moreover, in the above-described embodiment, the case in which the position adjustment portion **34** which is the guide groove is applied as the position adjustment portion has been described. However, the invention is not limited thereto. For example, various configurations of mechanisms may be applied as long as they can move the shielding plate **35** in the scanning direction of the laser beam LB.

(62) Further, in the above-described embodiment, as the angle adjustment portion, the angle adjustment portion **36** is provided which rotates on the rotating portion **37a** provided at the base end of the shielding plate **35** to move the shielding plate **35** along the guide portion **37b**, thereby inclining the shielding plate **35**. However, the invention is not limited thereto. For example, only the rotating portion **37a** may be provided to rotate the shielding plate **35** such that the angle can be adjusted, or only the guide portion **37b** may be provided to incline the shielding plate **35** such that the angle can be adjusted.

EXAMPLES

(63) Next, examples will be described. Here, first, the laser power of the laser beam LB or the like was defined, and the angle θ_0 formed between the reference position ($\theta = 0(^{\circ})$) in the plane mirror of the polygon mirror **10** and the boundary with an adjacent plane mirror was calculated. In addition, the critical angle θ_c was calculated from the above-mentioned Expression (1).

(64) In this case, when the laser power of the laser beam LB was 1000 (W), the radius ϕ of the laser

beam LB was 6 (mm), the number of plane mirrors N in the polygon mirror **10** was eight, and the circumscribed radius R of the polygon mirror **10** was 140 (mm), the angle θ_0 was $22.5(^{\circ})$, and the critical angle θ_c was $19.9(^{\circ})$.

(65) When the distance d between the point P and the point P1 was calculated from the above-mentioned Expression (2) assuming that the distance L1 from the plane mirror of the polygon mirror **10** to the lens **12** which was the condensing optical system was 50 (mm), the distance L2 from the lens **12** to the height of the point P of the shielding plate **35** was 150 (mm), and the focal length f of the lens **12** was 200 (mm), the distance d was 164.7 (mm).

(66) From the above, the position adjustment portion **34** can adjust the position of the shielding plate **35** in the scanning direction of the laser beam LB on the basis of the calculation result of the distance d.

(67) Then, the height difference Lp0 between the point P and the point P0 when the shielding plate **35** was inclined at the angle ψ was calculated, and the relationship between the angle ψ of the shielding plate **35** and the height difference Lp0 was investigated. The results shown in FIG. 6 were obtained. In FIG. 6, the horizontal axis indicates the angle $\psi(^{\circ})$ of the shielding plate **35**, and the vertical axis indicates the height difference Lp0 (mm) between the point P and the point P0 when the shielding plate **35** is inclined at the angle ψ .

(68) Here, as described in the embodiment, it is desirable that the angle ψ of the shielding plate **35** with respect to the surface of the steel sheet **20** is adjusted in the range of $2\theta_c < \psi \leq 90(^{\circ})$ and that $Lp0 < L2$ is satisfied as a constraint condition. Therefore, it was confirmed from FIG. 5 that the minimum angle $2\theta_c(^{\circ})$ of the angle ψ of the shielding plate **35** was about $40(^{\circ})$ and the maximum angle of the angle ψ of the shielding plate **35** was about $70(^{\circ})$ from the constraint condition.

(69) In addition, it can be seen that the optimum range of the angle ψ of the shielding plate **35** is from $40(^{\circ})$ to $70(^{\circ})$ and the most desirable angle ψ is $55(^{\circ})$ which is the central angle. From the above, the angle adjustment portion **36** can adjust the angle ψ of the shielding plate **35** on the basis of the above-mentioned calculation results.

INDUSTRIAL APPLICABILITY

(70) According to the invention, the shielding plate is inclined to reduce the damage of the shielding plate which occurs when the shielding plate blocks the laser beam. Therefore, it is possible to provide a groove processing device and a groove processing method that suppress the contamination of optical components and achieve uniform groove processing and groove depth. Therefore, the invention has extremely high industrial applicability.

BRIEF DESCRIPTION OF THE REFERENCE SYMBOLS

(71) **10** Polygon mirror **11** Light source device **12** Lens **20** Steel sheet **34** Position adjustment portion **35** Shielding plate **36** Angle adjustment portion **100** Groove processing device **101, 102** Plane mirror LB Laser beam

Claims

1. A groove processing device that forms a groove in a surface of an object using laser beams, the groove processing device comprising: a light source device that outputs the laser beams; a polygon mirror that reflects the laser beams output from the light source device such that the laser beam is scanned on the object; a condensing optical system, including a condensing lens or a mirror, that is provided on an optical path of the laser beams reflected by the polygon mirror and focuses the laser beams; and a shielding plate that is provided between the condensing optical system and the object at a position which blocks some of the laser beams focused through the condensing optical system and blocks some of the laser beams, wherein, among the laser beams focused through the condensing optical system, some of the laser beams that are not blocked by the shielding plate form the groove in the surface of the object at a focus of the laser beams, the shielding plate is configured to be inclinable so that the side closer to the center of the scan is the focus side and the

- side closer to the outside of the scan is the condensing optical system side, and the shielding plate is provided between the condensing optical system and the object and a sheet surface on which the laser beam is irradiated in the shielding plate is rotated with respect to the surface of the object in a direction in which the angle of the shielding plate changes with respect to the surface of the object so as to block the laser beams that do not form the groove.
2. The groove processing device according to claim 1, wherein, when an angle of the shielding plate with respect to the surface of the object is w and a critical angle which is a maximum angle at which the laser beam falls within one plane mirror of the polygon mirror is $\theta_c(^{\circ})$, the angle ψ of the shielding plate is inclined in a range of $2\theta_c < \psi \leq 90 (^{\circ})$.
 3. The groove processing device according to claim 2, wherein, assuming that a position where a perpendicular line is drawn from a rotation axis of the polygon mirror to the plane mirror of the polygon mirror is a reference position, an angle formed between a boundary between two adjacent plane mirrors of the polygon mirror and the reference position is $\theta_0 (^{\circ})$, a position where the shielding plate, which is inclined at the angle ψ , is irradiated with the laser beam reflected by the polygon mirror at an angle of $2\theta_0 (^{\circ})$ when a rotation angle of the polygon mirror is $\theta_0(^{\circ})$ is a point P_0 , a position where the shielding plate, which is inclined at the angle ψ , is irradiated with the laser beam reflected by the polygon mirror at an angle of $2\theta_c(^{\circ})$ when the rotation angle of the polygon mirror is $\theta_c(^{\circ})$ is a point P , a height difference between the point P and the point P_0 is L_{p0} , and a distance from the condensing optical system to a height of the point P is L_2 , $L_{p0} < L_2$ is satisfied.
 4. The groove processing device according to any one of claims 1 to 3, further comprising: a position adjustment portion, including a guide groove that adjusts a position of the shielding plate in a scanning direction in which scanning is performed with the laser beams by the polygon mirror.
 5. The groove processing device according to claim 1, further comprising: a housing that has the shielding plate disposed in a lower portion, wherein the housing has an upper opening portion that is located on the optical path of the laser beams focused by the condensing optical system, and a colorless and transparent window plate that transmits the laser beams without absorbing or reflecting the laser beams is attached to the upper opening portion.
 6. A groove processing method that forms a groove in a surface of an object using laser beams, the groove processing method comprising: an output step of outputting the laser beams from a light source device; a reflection step of reflecting the laser beams output from the light source device such that the laser beam is scanned on the object by a polygon mirror; a condensing step of focusing the laser beams on the surface of the object using a condensing optical system, including a condensing lens or a mirror, that is provided on an optical path of the laser beams reflected by the polygon mirror; and a shielding step of blocking some of the laser beams using a shielding plate that is provided between the condensing optical system and the object at a position which blocks some of the laser beams focused through the condensing optical system, wherein, among the laser beams focused through the condensing optical system, some of the laser beams that are not blocked by the shielding plate form the groove in the surface of the object at a focus of the laser beams, the shielding plate is configured to be inclinable so that the side closer to the center of the scan is the focus side and the side closer to the outside of the scan is the condensing optical system side, and in the shielding step, the shielding plate is provided between the condensing optical system and the object and a sheet surface on which the laser beam is irradiated in the shielding plate is rotated with respect to the surface of the object in a direction in which the angle of the shielding plate changes with respect to the surface of the object so as to block the laser beams that do not form the groove.
 7. The groove processing method according to claim 6, wherein, in the shielding step, when an angle of the shielding plate with respect to the surface of the object is w and a critical angle which is a maximum angle at which the laser beam falls within one plane mirror of the polygon mirror is $\theta_c(^{\circ})$, the angle ψ of the shielding plate is inclined in a range of $2\theta_c < \psi \leq 90 (^{\circ})$.
 8. The groove processing method according to claim 7, wherein, in the shielding step, assuming that a position where a perpendicular line is drawn from a rotation axis of the polygon mirror to the

plane mirror of the polygon mirror is a reference position, an angle formed between a boundary between two adjacent plane mirrors of the polygon mirror and the reference position is $\theta_0(^{\circ})$, a position where the shielding plate, which is inclined at the angle ψ , is irradiated with the laser beam reflected by the polygon mirror at an angle of $2\theta_0(^{\circ})$ when a rotation angle of the polygon mirror is $\theta_0(^{\circ})$ is a point P_0 , a position where the shielding plate, which is inclined at the angle ψ , is irradiated with the laser beam reflected by the polygon mirror at an angle of $2\theta_c(^{\circ})$ when the rotation angle of the polygon mirror is $\theta_c(^{\circ})$ is a point P , a height difference between the point P and the point P_0 is L_{p0} , and a distance from the condensing optical system to a height of the point P is L_2 , $L_{p0} < L_2$ is satisfied.

9. The groove processing method according to any one of claims 6 to 8, further comprising: a shielding plate position adjustment step of adjusting a position of the shielding plate in a scanning direction in which scanning is performed with the laser beams by the polygon mirror.

10. The groove processing method according to claim 6, further comprising: a housing attachment step of attaching a colorless and transparent window plate that transmits the laser beams without absorbing or reflecting the laser beams to an upper opening portion of a housing that has the shielding plate disposed in a lower portion and has the upper opening portion which is located on the optical path of the laser beams focused by the condensing optical system.
